

Quantifying life history

Annie Schiffer

2026-07-21

Learning objectives

1. Understand how and why growth, survival, and fecundity trade off
2. Do sensitivity and elasticity analyses

review of last lecture: life tables and structured populations. How do we compare species with different life history strategies?

analyze structured populations by manipulating elements of the matrix (vital rates)

Tradeoffs

most basic tradeoff is the r - K theory

r-K

- r species - fast growth r , many offspring, low investment
- K species - slow growth, near carrying capacity, few offspring, high investment
- **caveats** - binary instead of spectrum, generalization *example, trees like oaks that are long-lived but produce loads of acorns with low investment per acorn defy this theory*

Demographic triangle

- growth, stasis, fecundity

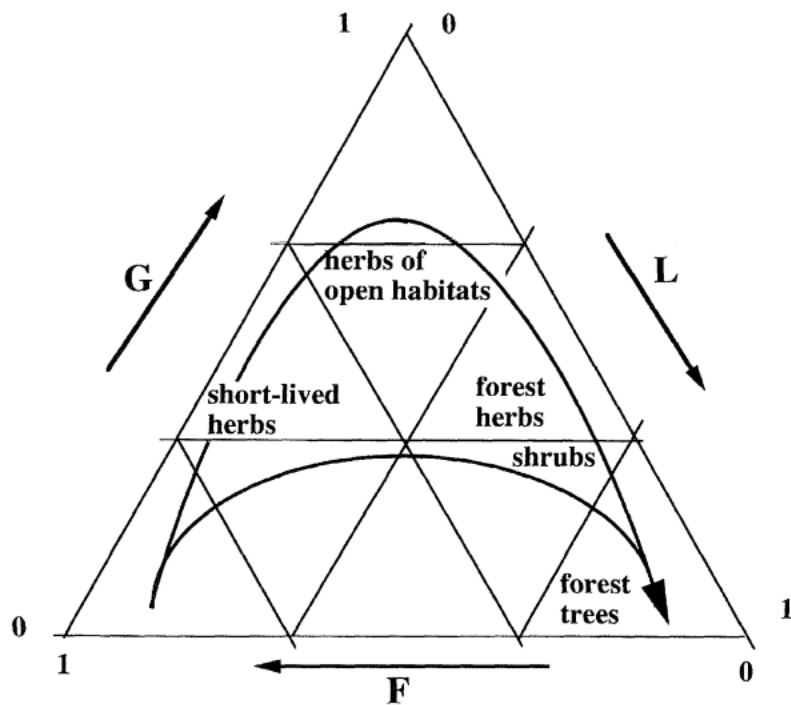


Figure 2. The successional trajectories suggested by the distribution of species in the demographic triangle in Fig. 1, after Silvertown & Franco (1993).

why can't you maximize growth, survival, and reproduction?

limited amount of energy and resources - need to decide how to allocate to growth, survival, reproduction

- can't optimize all three

Analyzing structured populations

(skip this?) Population projection

we can also project the abundances of each age in a given time step

- multiply: Leslie matrix, matrix of N at t
- product: matrix of N at t+1

quick refresher of matrix math

$$\begin{pmatrix} a & b & c \\ d & e & f \\ g & h & i \end{pmatrix} \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} ax + by + cz \\ dx + ey + fz \\ gx + hy + iz \end{pmatrix}$$

```
leslie <- c(0,0.2,0,0,0,0.25,100,0,0)
Nt <- c(1000, 200, 50)
```

```
leslie.mat <- matrix(leslie,nrow=3,ncol=3)
leslie.mat
```

```
##      [,1] [,2] [,3]
## [1,]  0.0 0.00 100
## [2,]  0.2 0.00  0
## [3,]  0.0 0.25  0
```

```
N.mat <- matrix(Nt,nrow=3,ncol=1)
N.mat
```

```
##      [,1]
## [1,] 1000
## [2,]  200
## [3,]   50
```

```
leslie.mat %*% N.mat
```

```
##      [,1]
## [1,] 5000
## [2,]  200
## [3,]   50
```

Elasticity analyses

let's say you have rare plant population data. population growth rate is less than 1, declining, but you don't know why. Is it survival or reproduction issue?

- quantify influence of vital rates on λ
- proportional change in growth rate
- relative importance/contribution of vital rates

not absolute change in population growth rate, use sensitivity analysis for that

Sensitivity analyses

let's say you could actually improve the survival of the population. Sensitivity analysis tells you how much the population growth rate would increase if you increased survival rate

- absolute change in λ with change in vital rates
- stage/age specific *increase survival of adults, juveniles, etc*
- repeatedly compute population growth rate